

```
-+-+  
X|O| 4 5 6  
-+-+  
X|O|X 7 8 9  
X has won the game!  
Thanks for playing!
```

Briefly, this program works by using a dictionary object to represent the nine spaces on a tic-tac-toe board. The dictionary's keys are the strings '1' through '9', and its values are the strings 'X', 'O', or ' '. The numbered spaces are in the same arrangement as a phone's keypad.

The functions in *tictactoe.py* do the following:

- The `main()` function contains the code that creates a new board data structure (stored in the `gameBoard` variable) and calls other functions in the program.
- The `getBlankBoard()` function returns a dictionary with the nine spaces set to ' ' for a blank board.
- The `getBoardStr()` function accepts a dictionary representing the board and returns a multiline string representation of the board that can be printed to the screen. This is what renders the tic-tac-toe board's text that the game displays.
- The `isValidSpace()` function returns `True` if it's passed a valid space number and that space is blank.
- The `isWinner()` function's parameters accept a board dictionary and either 'X' or 'O' to determine whether that player has three marks in a row on the board.
- The `isBoardFull()` function determines whether the board has no blank spaces, meaning the game has ended. The `updateBoard()` function's parameters accept a board dictionary, a space, and a player's X or O mark and updates the dictionary.

Notice that many of the functions accept the variable `board` as their first parameter. That means these functions are related to each other in that they all operate on a common data structure.